

RandModShift: Model and Data Privacy in Federated Inference

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Abstract—A key feature of federated learning is to ensure that the data used for distributed model learning stays private. While Differential Privacy (DP) has become a standard approach to mitigate data leakage, there are few solutions to ensure model privacy from an eavesdropper during learning. We extend here our prior work on ModShift, a method to increase model privacy via deterministic shifted model updates, to provide both model and data privacy. The crux of our approach, which we call RandModShift, is to introduce randomized shifts in model updates to ensure DP. The design strategy is based on posing the eavesdropper learning problem as an estimation problem, whose performance is governed by the Fisher Information Matrix (FIM) of the model updates. Our random shift design ensures that the FIM is singular and thus the eavesdropper experiences a provably hard learning problem. We derive technical conditions to ensure FIM singularity. We illustrate our approach with numerical results that show how RandModShift achieves a larger model shift and learning loss compared to a sole DP/noise injection scheme when both methods achieve the same DP levels.

Index Terms—Model privacy, federated learning, Fisher Information Matrix, estimation, distributed optimization.

I. INTRODUCTION

Federated learning [1]–[4] has emerged as a popular strategy for learning overarching data generation models from distributed, and potentially heterogeneous, data sets residing with clients. Local models are learned and shared with a global server; multiple learning iterations between the global server and clients lead to the learning of a global model at the server. This approach enables global learning without explicitly sharing the data sets and can reduce communication costs. A challenge with many federated learning methods is the leakage of information about the data sets [5], [6]. To this end additional methods for privacy to work with federated learning have emerged [7]–[9]. In particular, Differential Privacy (DP) [10]–[12] as a form of noise injection has emerged as a popular solution [13], [14].

Model and data privacy are both needed for modern applications. Our use of the term *model privacy* differs from that used in the context of *model extraction* [15]–[17]. Therein, there exists an already learned model that an adversary endeavors reverse engineering from input-output pairs. Instead, we seek here to ensure privacy from an eavesdropper overhearing model updates over a communication channel to construct

the final model. The limited work on this problem [18]–[20] suffers from high complexity or impractical assumptions.

In previous work, we have proposed a model privacy scheme called *ModShift* [21], which is based on driving the Fisher Information Matrix (FIM) for the eavesdropper’s learning problem to singularity to introduce significant model errors. We generalized the approach in [22] to prevent the eavesdropper from using the correlation of updates to aid their estimation. In the present work, we consider the problem of ensuring both model and data privacy. This problem is partially addressed in [18], [19] but without offering a computationally lightweight and communication efficient schemes. The key difference with our prior work [21] is that we now design *randomized* shifts of the transmitted model updates. We therefore call our scheme *RandModShift*: we design randomized shifts that achieve DP of the data at each client in addition to model privacy. Our approach does not make any assumptions about the underlying learning problem [18] or the number of compromised links [20]. Furthermore, RandModShift is computationally lightweight and communication efficient.

The main contributions of the present work are as follows.

- 1) We generalize our prior results to design randomized model shifts to achieve model privacy in distributed learning. The needed conditions on the randomized shifts to ensure a singular FIM for an eavesdropper are derived.
- 2) A joint data-model privacy which injects DP noise is proposed. The conditions to achieve DP data privacy metrics as well as model privacy from the eavesdroppers FIM are derived.
- 3) The FIM singularity condition is used to characterize a family of random shifts that joint provide model and data privacy.
- 4) Numerical results show, that for the same level of DP, our proposed scheme leads Eve to a model loss that is 2 – 3 orders of magnitude higher than a noise injection scheme and prior work [20].

The paper is organized as follows. We present our system model in Section II and review key concepts of DP as they relate to our problem. We present our proposed scheme, *RandModShift*, in Section IV. We present numerical results

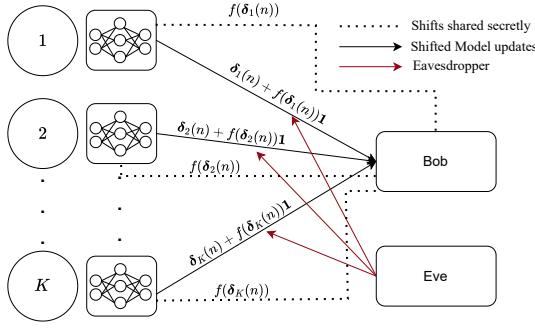


Fig. 1. System model for distributed optimization, model shift and eavesdropping.

in Section V and offer concluding remarks in Section VI.

II. SYSTEM MODEL

Notation. We write vectors and matrices in boldface. Random quantities are underlined. We denote the probability density function of random variable $\underline{x}$, evaluated at realization x , and parametrized by y by $p_{\underline{x}}(x; y)$. For a positive integer K , we define $[K] = \{1, 2, \dots, K\}$.

As illustrated in Fig. 1, we consider a network of K agents (Alices) that communicate with a central server (Bob) to collectively learn a global model $\mathbf{w}^* \in \mathbb{R}^d$ using a method such as FedAvg [23]. Each agent k has a local dataset of size m_k with labeled pairs $\mathbf{D}_k = \{(\mathbf{x}_{k,i}, y_{k,i}) : i \in [m_k]\}$, where $(\mathbf{x}_{k,i}, y_{k,i})$ represents the feature vector and associated label for the i th data point at agent k . Define $m = \sum_{k=1}^K m_k$. To learn $\mathbf{w}^*$, Bob minimizes the empirical loss over the aggregated data:

$$F(\mathbf{w}) = \frac{1}{m} \sum_{k=1}^K \sum_{i=1}^{m_k} \ell(\mathbf{w}, \mathbf{x}_{k,i}, y_{k,i}), \quad (1)$$

where $\ell(\mathbf{w}, \mathbf{x}_{k,i}, y_{k,i})$ is the loss associated to the prediction of $y_{k,i}$ from $\mathbf{x}_{k,i}$ using model $\mathbf{w}$. A standard approach to compute $\mathbf{w}^*$ is to use an iterative method such as gradient descent (either stochastic or deterministic). For a model $\mathbf{w}$, the k th average gradient computed on dataset $\mathbf{D}_k$ is given by

$$\mathbf{g}_k(\mathbf{w}, \mathbf{D}_k) = \frac{1}{m_k} \sum_{i=1}^{m_k} \nabla \ell(\mathbf{w}, \mathbf{x}_{k,i}, y_{k,i}). \quad (2)$$

Each agent k performs R local epochs, each corresponding to an iteration of gradient descent

$$\mathbf{w}_{k,r+1} = \mathbf{w}_{k,r} - \eta \hat{\mathbf{g}}_k(\mathbf{w}_{k,r}, \mathbf{D}_k), \quad (3)$$

where $\hat{\mathbf{g}}_k$ is either $\mathbf{g}_k$ or an estimate thereof and $\eta > 0$ is a predefined learning rate. After R local epochs, the agents send the difference between their initial weights and $\mathbf{w}_{k,R}$ to Bob, who computes a weighted average and sends the updated model back for the next round of agent updates. We define the message sent during the n th round of communication between agents and Bob as

$$\boldsymbol{\alpha}_k(n) = \mathbf{w}_{k,R}(n) - \mathbf{w}(n). \quad (4)$$

The pseudocode of the complete federated learning process is given in Algorithm 1. **Anand: I don't think you need the pseudocode for regular federated GS. If you need pseudocode at all it should be for the proposed method.**

Algorithm 1 The K clients are indexed by k ; R is the number of local epochs and η is the learning rate.

Server executes:

Initialize $\mathbf{w}(0)$

for each round $n = 1, 2, \dots, N$ **do**

for each client k **in parallel do**

$\mathbf{w}_{k,0}(n) \leftarrow \mathbf{w}(n)$

for each local epoch r from 0 to $R - 1$ **do**

$\mathbf{w}_{k,r+1}(n) = \mathbf{w}_{k,r}(n) - \eta \hat{\mathbf{g}}_k(\mathbf{w}_{k,r}(n), \mathbf{D}_k)$

end for

$\boldsymbol{\alpha}_k(n) \leftarrow \mathbf{w}_{k,R}(n) - \mathbf{w}(n)$

end for

$\mathbf{w}(n+1) \leftarrow \mathbf{w}(n) + \sum_{k=1}^K \frac{m_k}{m} \boldsymbol{\alpha}_k(n)$

end for

Finally, we assume that there exists an adversary (Eve) who can eavesdrop the links between the agents and the server. There are at least two threats to consider in this setting. First, if Eve knows $\mathbf{w}(0)$ and observes the gradient updates $\{\boldsymbol{\alpha}_k(n) : k \in [K]\}$, she can track the model updates and infer the model as it is being learned: we call this a *model privacy violation*. Second, if Eve observes the gradient updates, she may be able to reconstruct some of the training points or identify if a particular training point $\{(\mathbf{x}_{k,i}, y_{k,i})\}$ is in one of the data sets: we call this a *data privacy violation*. In the present, paper we are interested in mitigating both types of privacy violations.

Anand: Just a thought,

III. DIFFERENTIAL PRIVACY

Differential Privacy (DP) [24] has perhaps become the most popular method to ensure data privacy in machine learning. For completeness, we provide here some background, although we primarily focus on the Gaussian mechanism described next. Two datasets $\mathbf{D}$ and $\mathbf{D}'$ are called *neighboring* if they differ in a single data point.

Definition 1 ((ϵ, δ) -DP). A stochastic map $\mathcal{M}$ is (ϵ, δ) -differentially private $((\epsilon, \delta)$ -DP) if for all datasets $\mathbf{D}, \mathbf{D}'$, differing in one element, and for any measurable set $\mathcal{S}$,

$$\mathbb{P}(\mathcal{M}(\mathbf{D}) \in \mathcal{S}) \leq e^\epsilon \mathbb{P}(\mathcal{M}(\mathbf{D}') \in \mathcal{S}) + \delta. \quad (5)$$

This privacy definition has a hypothesis testing interpretation [25], [26]: an adversary observing the output of $\mathcal{M}$ wishes to test whether the input was $\mathbf{D}$ or $\mathbf{D}'$. If $\mathcal{M}$ is (ϵ, δ) -DP the probability of false alarm P_{FA} and missed detection P_{MD} are lower bounded as follows:

$$P_{\text{FA}} + e^\epsilon P_{\text{MD}} \geq 1 - \delta, \quad (6)$$

$$e^\epsilon P_{\text{FA}} + P_{\text{MD}} \geq 1 - \delta. \quad (7)$$

Thus, low values of (ϵ, δ) imply that the outputs of two datasets that differ by a single entry are close to each other. In other words, it is difficult to tell whether a particular individual record is in the original dataset.

A map $\mathcal{M}$ for ensuring differential privacy is called a *mechanism*. We focus on Gaussian mechanism, which makes a function $f(\mathbf{D})$ differentially private by adding Gaussian noise. Specifically, the Gaussian mechanism for approximating f with differential privacy is $\mathcal{M}(\mathbf{D}) = f(\mathbf{D}) + \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$ and τ^2 is a function of the sensitivity Δ of f defined as follows.

Definition 2 ($\mathcal{L}_2$ -sensitivity). *The $\mathcal{L}_2$ -sensitivity of a vector-valued function $f(\mathbf{D})$ is $\Delta = \max_{\mathbf{D}, \mathbf{D}'} \|f(\mathbf{D}) - f(\mathbf{D}')\|_2$, where $\mathbf{D}, \mathbf{D}'$ are neighboring datasets.*

One can interpret the (ϵ, δ) -DP as measuring the distance between the two distributions $\mathcal{N}(f(\mathbf{D}), \tau^2 \mathbf{I})$ and $\mathcal{N}(f(\mathbf{D}'), \tau^2 \mathbf{I})$ using an f -divergence known as the Hockey-Stick Divergence (see the recent tutorial [12]). The variance τ^2 can be translated into a whole curve of (ϵ, δ) values: δ is related to the tail probability and ϵ to a log-likelihood ratio. Specifically, the following result holds.

Proposition 1 ([27]). *Let $f : \mathcal{D} \rightarrow \mathbb{R}^d$ be an arbitrary differentiable function with $\mathcal{L}_2$ -sensitivity Δ . The Gaussian mechanism with parameter τ adds noise $\mathcal{N}(0, \tau^2 \mathbf{I})$ to the output and satisfies (ϵ, δ) -differential privacy for $\epsilon \in (0, 1)$ if $\tau \geq \frac{\Delta}{\epsilon} \sqrt{2 \log(\frac{1.25}{\delta})}$.*

While many papers use the τ given in Proposition 1 (valid for $\epsilon < 1$, a formula relating τ and (ϵ, δ) valid for any ϵ is available [28]).

Anand: If we want more space we can remove Proposition 1 with the following: Typically, privacy loss can be tracked as a function of noise variances using Rényi-DP [29] and converted to a particular (ϵ, δ) pair.

Turning to our model training scenario, one popular method for guaranteeing (ϵ, δ) -DP in iterative optimization is to add Gaussian noise to the gradients or updates $\alpha_k(n)$. In order to bound the $\mathcal{L}_2$ -sensitivity Δ , the model updates are clipped before adding noise:

$$\alpha_k^c(n) = \text{clip}(\alpha_k(n), c) = \alpha_k(n) \cdot \min \left\{ 1, \frac{c}{\|\alpha_k(n)\|} \right\}. \quad (8)$$

Thus, the update after these operations is given by

$$\underline{\alpha}_k^{\text{DP}}(n) = \alpha_k^c(n) + \underline{\theta}_k(n). \quad (9)$$

where $\underline{\theta}_k(n) \sim \mathcal{N}(0, \tau^2 \mathbf{I}_d)$.¹

Note that the Gaussian noise mechanism ensures data privacy from both Bob and Eve but does not provide model privacy against Eve since Eve receives the same noisy updates as Bob. Therefore, with only DP, Bob and Eve learn the same model and additional processing is needed to provide model privacy from Eve. To compare various model privacy schemes,

¹Clipping slows the rate of convergence but does not increase the rank of the derived FIM.

we will provide the same level of data privacy by adding $\underline{\theta}_k(n)$ to the model updates and then evaluate the effectiveness of the model privacy schemes. Rather than select an (ϵ, δ) pair, we will select τ as per Proposition 1.

IV. PROPOSED SCHEME: RANDMODSHIFT

Our previous approaches ModShift [21] and Block ModShift [22] are *shift-based model* privacy preserving schemes, they do not analyze data privacy or propose a method to merge their approaches with existing data privacy methods. One approach to merging these two kinds of protections would be “Shift-then-Noise”: apply the original update-shifting strategy and then add Gaussian noise to provide DP. The drawback of this approach is that the $\mathcal{L}_2$ -sensitivity of the shifted model update may be larger than the unshifted update and therefore require more noise for the same level of privacy. We propose the reverse, “Noise-then-Shift”: add noise to the original update and then shift the noisy update. An advantage of this approach is the DP guarantees are inherited from standard results: any subsequent shifting does not increase the privacy risk by the post-processing invariance of differential privacy. The complication is that the effective shift now becomes a random vector, unlike the deterministic settings previously studied [21], [22]. We call our new scheme RandModShift: in the sequel we define the scheme more formally and provide an analysis to show it preserves model privacy.

A. RandModShift

Each agent k has an update $\alpha_k(n)$ known to them and generates Gaussian noise $\underline{\theta}_k(n) \sim \mathcal{N}(0, \tau^2 \mathbf{I})$. The differentially private update for agent k is given in (9). Each agent chooses a direction $\mathbf{u}_k(n)$ with $\|\mathbf{u}_k(n)\|_2 = 1$ which is known to all parties. Corresponding to each $\mathbf{u}_k(n)$, a shift vector $\gamma_k(n)$ needs to be designed. The agent computes a scalar *shift*:

$$\zeta_k(n) = \gamma_k(n)^T \underline{\alpha}_k^{\text{DP}}(n) \quad (10)$$

$$= \gamma_k(n)^T (\alpha_k^c(n) + \underline{\theta}_k(n)) \quad (11)$$

We assume the agent can send this scalar shift to Bob over the secure channel so that only Bob and the agent know the realization of $\zeta_k(n)$. The parameters $\mathbf{u}_k(n)$ and $\gamma_k(n)$ are known to everyone. We note that Bob does not need to know $\gamma_k(n)$ to cancel the shift.

Each agent $k \in [K]$ shares the following shifted update to Bob and Eve:

$$\underline{\mathbf{x}}_k(n) = \underline{\alpha}_k^{\text{DP}}(n) + \zeta_k(n) \mathbf{u}_k(n). \quad (12)$$

$$= \alpha_k^c(n) + \underline{\theta}_k(n) + \gamma_k(n)^T (\alpha_k^c(n) + \underline{\theta}_k(n)) \mathbf{u}_k(n). \quad (13)$$

Bob can compute $\zeta_k(n) \mathbf{u}_k(n)$ and subtract it, leaving him with

$$\underline{\mathbf{y}}_k^{\text{B}}(n) = \alpha_k^c(n) + \underline{\theta}_k(n) \quad k \in [K]. \quad (14)$$

Bob can apply standard noisy gradient descent: the convergence rate is dictated by the privacy parameters/level of the noise τ^2 . By contrast, Eve has

$$\underline{\mathbf{y}}_k^{\text{E}}(n) = \underline{\mathbf{x}}_k(n) \quad k \in [K]. \quad (15)$$

Our goal is to develop conditions on $\gamma_k(n)$ such that the distribution of the random variable $\zeta_k(n)$ can provide similar guarantees for model privacy as the deterministic strategies in ModShift [21] and Block ModShift [22].

B. Understanding Eve's errors

We analyze Eve's error under in the case where she naively² applies the same gradient update rule:

$$\mathbf{w}(n+1)^E = \mathbf{w}(n)^E + \sum_{k=1}^K \frac{m_k}{m} \mathbf{y}_k^E(n). \quad (16)$$

We will develop designs that will drive Eve's FIM [30] to singularity. We assume the following:

Assumption 1. Eve treats the model updates $\alpha_k(n)$ as deterministic unknowns.

Note that while the above assumption is made for analysis, the simulations in Section V are not constrained. In the absence of any priors, Eve poses an estimation problem for the deterministically modeled $\alpha_k^c(n)$. For each user k and each round n from Equation (13). We can compute a set of FIMs for each k and n . The optimal shift parameters are determined via the following optimization:

$$\mathcal{P} : p_{\zeta_k(n)}(\cdot) = \arg \min_{p_{\zeta_k(n)}(\cdot)} |\det(\mathbf{J}(\alpha_k^c(n)))| \quad (17)$$

where $\mathbf{J}(\alpha_k^c(n))$ is the Fisher Information Matrix of the clipped differences sent by agent k in round n and is given by

$$\mathbf{J}(\alpha_k(n)) = -\mathbb{E} \left[\nabla_{\alpha_k^c(n)}^2 \log p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) \right], \quad (18)$$

where $p_{\mathbf{y}_k^E(n)}(\cdot; \alpha_k^c(n))$ denotes the probability density function of $\mathbf{y}_k^E(n)$, parameterized by $\alpha_k(n)$. Our next task is to characterize the FIM for Eve. We shall see that we can determine constraints on the shifts to ensure that her FIM is singular.

Proposition 2. $\mathbf{J}(\alpha_k^c(n))$ for the signal model in Equation (13) is singular when

$$\gamma_k(n)^T \mathbf{u}_k(n) = -1 \quad (19)$$

Proof. See Appendix. $\square$

We note that if $\tau = 0$ (see Proposition 1), the scheme reduces to ModShift. The shift is computed and added before transmission and the shift structure is independent of the channel statistics, so that our scheme is independent of the channel model.

V. SIMULATION RESULTS

We consider a linear regression problem on a synthetic dataset in a federated learning setup with 100 independent agents communicating with Bob. The data vectors $\mathbf{x}_{k,i}$ are of dimension $d = 60$ and are Gaussian distributed with mean 0 and identity covariance matrix. The weight vector $\mathbf{w} = [1, 2, \dots, d]^T$ is fixed and labels for each data vector are

generated as $y_{k,i} = \mathbf{w}^T \mathbf{x}_{k,i} + n_{k,i}$ where $n_{k,i}$ is a zero-mean Gaussian random variable with variance 1 for all k, i . Each agent has a dataset of size 1,000. We use the mean square error loss function for which $\ell(\mathbf{w}, \mathbf{x}_{k,i}, y_{k,i})$ in Equation (1) is given by

$$\ell(\mathbf{w}, \mathbf{x}_{k,i}, y_{k,i}) = (\mathbf{w}(n)^T \mathbf{x}_{k,i} - y_{k,i})^2. \quad (20)$$

The learning rate is set to $\eta = 0.005$. We set the initial weights $\mathbf{w}^{Bob}(0) = \mathbf{w}^{Eve}(0) = \mathbf{0}$ where $\mathbf{0} \in \mathbb{R}^d$ is the zero vector and $\frac{\sigma_B^2}{(h_k^B(n))^2} = \frac{\sigma_E^2}{(h_k^E(n))^2} = 0.1 \quad \forall k, n$.

For differential privacy, we use the additive Gaussian mechanism discussed in Section III. We set the clipping parameter $C = 1$ and the standard deviation of the noise being added to $\tau = 0.5$. Therefore, rather than choose ϵ and δ , we set $\tau = 0.5$ which corresponds to multiple (ϵ, δ) pairs. For all comparison schemes, the same data privacy guarantee is provided and we compare each scheme's ability to degrade Eve's model as measured by Eve's final loss.

We compare RandModShift against two schemes: Noise Injection and SingleLink [20]. For all three schemes, we consider a "with DP" approach (labeled "scheme/DP") and a "without DP" (labeled "scheme") approach to compare the effect of DP on model privacy, as measured using Eve's loss under the scheme. For "with DP," Eve estimates the updates from

$$\mathbf{y}_k^E(n) = \alpha_k^{DP}(n) + \mathbf{s}_k(n), \quad (21)$$

where

$$\mathbf{s}_k(n) \sim \begin{cases} \gamma_k(n)^T \alpha_k^{DP}(n) \mathbf{u}_k(n), & \text{RandModShift/DP,} \\ \mathcal{N}(0, \beta^2 \mathbf{I}), & \text{Noise Injection/DP,} \\ 0, & \text{SingleLink/DP.} \end{cases} \quad (22)$$

For the "without DP" case, Eve estimates updates from

$$\mathbf{y}_k^E(n) = \alpha_k(n) + \mathbf{s}_k(n), \quad (23)$$

where $\mathbf{s}_k(n)$ remains the same for Noise Injection and SingleLink but for RandModShift, $\mathbf{s}_k(n) = \gamma_k(n)^T \alpha_k(n) \mathbf{u}_k(n)$. In both "with DP" and "without DP" cases, agents share a scalar over the secret channel in the case of RandModShift and the noise vector in the case of Noise Injection.

For RandModShift, we set the shift vector $\mathbf{u}_k(n) = \mathbf{1}$ and use the max scheme. With $\alpha = \arg \max_i |\alpha_{k,i}(n)|$, we have

$$\gamma_{k,i}(n) = \begin{cases} -1, & i = \alpha, \\ 0, & i \neq \alpha. \end{cases}$$

Figure 2 shows the loss as a function of the iterations for Bob and Eve under different schemes with and without differential privacy. For the noise injection scheme, we set $\beta = 1$. We note that both Bob and Eve's convergence rate is affected when DP noise is added, which is expected because of clipping. RandModShift leads Eve to a worse loss with DP noise because of the additional noise as well as the randomness introduced in the shift. While the noise injection scheme is initially affected by DP noise, eventually, the noise injection

²This simplifying assumption allows for an easier exposition of the results.

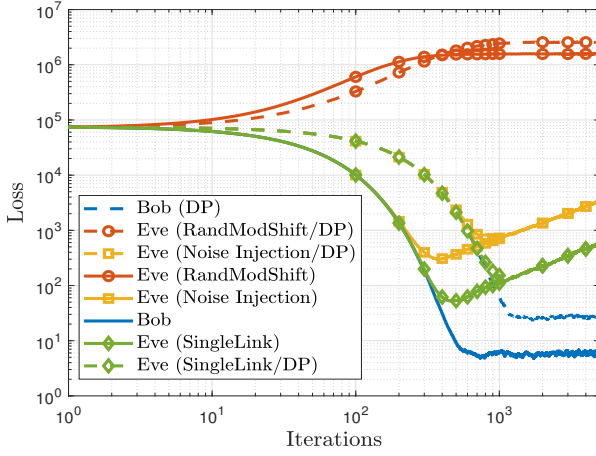


Fig. 2. Comparison of model privacy schemes with and without differential privacy guarantee.

scheme has the same performance with and without DP noise. The same can be seen in the case of SingleLink. All schemes have the same DP guarantee but RandModShift leads Eve to a loss that is 2 – 3 orders of magnitude more than the noise injection scheme and SingleLink. Moreover, a drawback of the noise injection scheme is that each agent must share a d -dimensional vector over the secret channel with Bob.

VI. CONCLUSIONS

We introduced a random shift framework to provide and facilitate the analysis of a joint data-model privacy preserving scheme against an eavesdropper in a federated learning environment. We treated model learning as a parameter estimation problem. This approach allowed us to derive conditions on under which random shifts make the estimation problem provably hard for Eve. We observe that for a given DP guarantee, our approach achieves a higher model shift than prior work [20] and noise addition scheme while requiring smaller usage of the secret channel between Bob and each agent. This work was motivated by scenarios in which an eavesdropper in a wireless system can intercept messages from individual agents to a central aggregator (Bob). If we can model the channels to Bob and to Eve as noisy, the FIM analysis provides lower bounds for estimation that can hold for a wide class of noise distributions. The optimization of the shift design and overall privacy accounting are deferred to future work.

VII. ACKNOWLEDGEMENTS

This work has been funded in part by one or more of the following grants: ONR N00014-22-1-2363, ARO W911NF2410094, NSF CCF-2311653, ARO W911NF1910269, and NSF DBI-2412522. This work was supported in part by the National Science Foundation (NSF) under grants 2148400, 2148104, 2148313 as part of the Resilient & Intelligent NextG Systems (RINGS) program.2148313

APPENDIX

PROOF OF PROPOSITION 2

Proof. We find conditions on the pdf of $\zeta_k(n)$ under which $\mathbf{J}(\alpha_k^c(n))$ singular. Consider

$$\mathbf{u}_k(n)^T \mathbf{J}(\alpha_k^c(n)) \mathbf{u}_k(n) \quad (24)$$

$$= \mathbb{E} \left[\mathbf{u}_k(n)^T \nabla_{\alpha_k^c(n)} \log \left(p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) \right) \right]$$

$$\nabla_{\alpha_k^c(n)} \log \left(p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) \right)^T \mathbf{u}_k(n) \quad (25)$$

$$= \mathbb{E} \left[\left(\mathbf{u}_k(n)^T \nabla_{\alpha_k^c(n)} \log \left(p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) \right) \right)^2 \right] \quad (26)$$

Setting the above to 0, we have

$$\mathbf{u}_k(n)^T \nabla_{\alpha_k^c(n)} \log \left(p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) \right) = 0. \quad (27)$$

Thus,

$$p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) = p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n) + t\mathbf{u}_k(n)) \quad \forall t. \quad (28)$$

$$p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n)) = \int_{\mathbb{R}} p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \zeta, \alpha_k^c(n)) p_{\zeta_k(n)}(\zeta; \alpha_k^c(n)) d\zeta \quad (29)$$

$$= \frac{1}{(2\pi\tau\|\gamma_k(n)\|^2)^{\frac{1}{2}}} \frac{1}{(2\pi\tau^2)^{\frac{d}{2}}} \int_{\mathbb{R}} e^{-\frac{\|(\mathbf{y} - (\alpha_k^c(n) + \zeta\mathbf{u}_k(n)))\|^2}{2\tau^2}} e^{-\frac{(\zeta - \gamma_k(n)^T \alpha_k^c(n))^2}{2\tau\|\gamma_k(n)\|^2}} d\zeta. \quad (30)$$

Similarly,

$$p_{\mathbf{y}_k^E(n)}(\mathbf{y}; \alpha_k^c(n) + t\mathbf{u}_k(n)) \quad (31)$$

$$= \frac{1}{(2\pi\tau^2)^{\frac{d}{2}}} \int_{\mathbb{R}} e^{-\frac{\|(\mathbf{y}_k^E(n) - (\alpha_k^c(n) + (\zeta + t)\mathbf{u}_k(n)))\|^2}{2\tau^2}} p_{\zeta_k(n)}(\zeta; \alpha_k^c(n) + t\mathbf{u}_k(n)) d\zeta, \quad (32)$$

$$= \frac{1}{(2\pi\tau^2\|\gamma_k(n)\|^2)^{\frac{1}{2}}} \frac{1}{(2\pi\tau^2)^{\frac{d}{2}}} \int_{\mathbb{R}} e^{-\frac{\|(\mathbf{y} - (\alpha_k^c(n) + \zeta'\mathbf{u}_k(n)))\|^2}{2\tau^2}} e^{-\frac{(\zeta - t - \gamma_k(n)^T (\alpha_k^c(n) + t\mathbf{u}_k(n)))^2}{2\tau^2\|\gamma_k(n)\|^2}} d\zeta. \quad (33)$$

Thus,

$$-t - \gamma_k(n)^T (\alpha_k^c(n) + t\mathbf{u}_k(n)) = -\gamma_k(n)^T \alpha_k^c(n), \quad (34)$$

$$\implies \gamma_k(n)^T \mathbf{u}_k(n) = -1 \quad (35)$$

□

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